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MODULE NAME : PROGRAMMING

MODULE CODE : PROG6212

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Contract Monthly Claim System - Documentation

Project Overview

The Contract Monthly Claim System is an [ASP.NET](#) Core MVC web application designed to streamline the submission, review, and tracking of monthly claims for university lecturers. The system caters to separate workflows for lecturers, program coordinators, and academic managers with suitable role-based access and approval flows.

System Architecture

Technology Stack

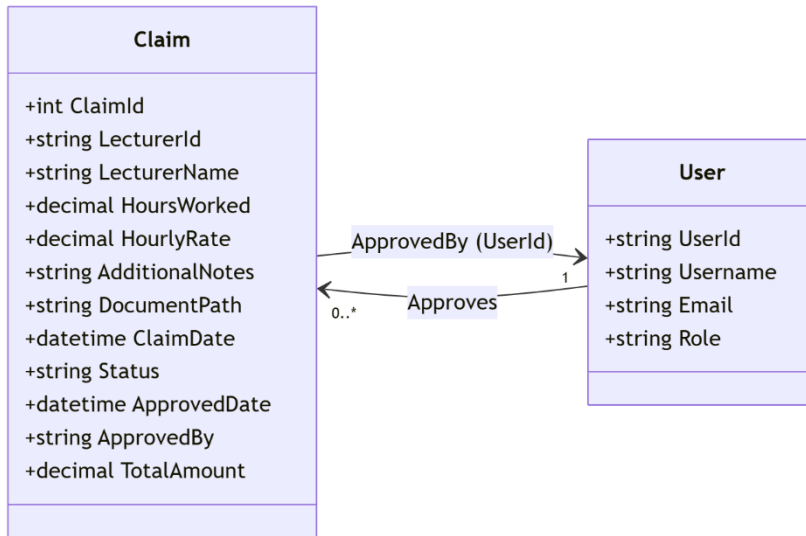
- **framework:** ASP.NET Core 6.0 MVC
- **Front end:** Bootstrap 5.1.3, Font Awesome 6.0
- **Testing:** xUnit, Moq
- **File Handling:** IFormFile with security restrictions

Design Patterns

- **MVC Pattern:** Clear separation of concerns between Models, Views, and Controllers
- **Repository Pattern:** Data access abstraction (implemented using static list for prototype)
- **Dependency Injection:** Built-in IoC container for service management

Database Design

UML Class Diagram



Entity Relationships

- User to Claim:** One-to-Many relationship (Single user for many claims)
- Status Workflow:** Pending → Approved/Rejected → Settled

Core Components

1. Models

Claim Model

C sharp

```

public class Claim
{
    // Primary key and identification
    public int ClaimId { get; set; }
    public string LecturerId { get; set; }
    public string LecturerName { get; set; }

    // Financial details
    public int HoursWorked { get; set; }
    public decimal HourlyRate { get; set; }
  
```

```
public decimal TotalAmount => HoursWorked * HourlyRate;
```

```
// Supporting information
```

```
public string AdditionalNotes { get; set; }
```

```
public string DocumentPath { get; set; }
```

```
// Status tracking
```

```
public DateTime ClaimDate { get; set; }
```

```
public ClaimStatus Status { get; set; }
```

```
public DateTime? ApprovedDate { get; set; }
```

```
public string ApprovedBy { get; set; }
```

```
// File upload (not persisted)
```

```
public IFormFile SupportingDocument { get; set; }
```

```
}
```

Important Features:

- Data Validation:** Range and string length validation attributes
- Computed Properties:** Automatic calculation of TotalAmount
- Status Enum:** Type-safe status management
- File Handling:** Secure document upload with type restrictions

2. Controllers

ClaimsController

Processes all claim operations with action methods for:

- Submit:** Submits a claim with file upload
- Pending:** Approval queue for coordinators/managers
- Approve/Reject:** Status update operations
- Track:** Claim status tracking for lecturers

Key Implementation Details:

File Upload Security

C sharp

// Secure file handling with size and type restrictions

```
var uploadsFolder = Path.Combine(_environment.WebRootPath, "documents");
```

```
var fileName = Guid.NewGuid().ToString() + Path.GetExtension(file.FileName);
```

- **Distinct Filenames:** Prevents conflict due to GUID-based naming
- **Directory Security:** Isolated documents folder
- **File Type Validation:** Limited to common document types

Status Management

C sharp

```
public IActionResult Approve(int id)
{
    var claim = _claims.FirstOrDefault(c => c.ClaimId == id);
    claim.Status = ClaimStatus.Approved;
    claim.ApprovedDate = DateTime.Now;
    claim.ApprovedBy = User.Identity.Name;
}
```

Real-time Updates: Instant status reflection

- **Audit Trail:** Records approver and timestamp

3. Views

Submit View

- **User-Friendly Form:** Simple Bootstrap layout with validation
- **File Upload:** Drag-and-drop friendly input
- **Real-time Validation:** Client-side and server-side validation

Pending View

- **Tabular Display:** Neat claim listing
- **Quick Actions:** Obvious approve/reject buttons
- **Document Access:** Direct links to supporting documents

Track View

- **Visual Status:** Color-coded badges for status variation
- **Progress Tracking:** Obvious indication of claim progress
- **Responsive Cards:** Mobile-friendly layout

Security Features

1. Input Validation

C sharp

[Required]

[Range(1, 200)]

```
public int HoursWorked { get; set; }
```

[Required]

[Range(0, 1000)]

```
public decimal HourlyRate { get; set; }
```

2. File Upload Security

- **Size Limit:** 5MB maximum file size
- **Type Restrictions:** PDF, DOCX, XLSX, JPG, PNG only
- **Secure Storage:** Files stored outside web root with GUID names

3. Anti-Forgery Protection

html

```
<form asp-action="Submit" method="post" enctype="multipart/form-data">
```

```
    @Html.AntiForgeryToken()
```

```
</form>
```

Testing Strategy

Unit Tests Coverage

C sharp

```
public class ClaimsControllerTests
```

```
{
```

```
    // Test scenarios covered:
```

```
    // - Form submission validation
```

```
    // - File upload handling
```

```
    // - Status update workflows
```

```
    // - Data retrieval operations
```

```
}
```

Test Categories:

1. **Controller Actions:** Ensure correct view return and redirects
2. **Business Logic:** Claim calculation and status change
3. **File Operations:** Upload and error handling
4. **Validation:** Model validation and input validation

User Workflows

Lecturer Workflow

1. **Login** → Access own dashboard
2. **Submit Claim** → Fill in form + attach documents
3. **Track Status** → Track approval status
4. **View History** → View previous claims

Coordinator/Manager Workflow

1. **Login** → Access approval dashboard
2. **Review Pending** → View claims awaiting decision
3. **Approve/Reject** → Decide with comments
4. **Monitor Queue** → Track total claim quantity

Error Handling

Graceful Error Management

C sharp

```
try
{
    // File operations
}
catch (Exception ex)
{
    // Log error and display user-friendly message
    TempData["ErrorMessage"] = "File upload failed. Please try again.";
}
```

User Feedback

- **Success Messages:** Green notifications for finished actions
- **Error Messages:** Red notifications with precise instructions
- **Validation Errors:** Highlighting per field errors

Configuration Settings

appsettings.json

```
json
{
  "FileUpload": {
    "MaxFileSize": 5242880,
    "AllowedExtensions": ".pdf,.docx,.xlsx,.jpg,.png"
  }
}
```

Assumptions and Constraints

Technical Assumptions

1. **Authentication:** External authentication system present
2. **User Roles:** Role-based access control implemented
3. **Storage:** File system storage sufficient for prototype
4. **Concurrency: Single-user operations assumed**

Business Constraints

1. **Approval Chain:** Two-level approval (Coordinator → Manager)
2. **Document Requirements:** Supporting documents are required
3. **Financial Limits:** Hourly rate and hours limits
4. **Audit Requirements:** Full audit trail kept

Future Enhancements

Planned Improvements

1. **Database Integration:** Static list replaced with Entity Framework
2. **Email Notifications:** Automated status notification
3. **Advanced Reporting:** Analytics and reporting dashboard
4. **Bulk Operations:** Batch approval and processing
5. **Native Mobile App**

Deployment Considerations

Infrastructure Requirements

- **Web Server:** IIS or Kestrel
- **Storage:** File system with backup strategy

- Security:** SSL certificate for production
- Monitoring:** Application performance monitoring

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